

Predicting Bike-Sharing Demand

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DATA SCIENCE PROJECT CANVAS

Title: Bike Sharing Demand Forecast

1. Problem Statement

Bike sharing systems are a means of renting bicycles where the process of obtaining membership, rental, and bike return is automated via a network of kiosk locations throughout a city. Using these systems, people are able rent a bike from a one location and return it to a different place on an as-needed basis. Currently, there are over 500 bike-sharing programs around the world.

2. Outcomes/Predictions

We are predicting bike **rental demand** in the Capital Bikeshare program in Washington, D.C. The dataset combine historical usage patterns with weather data

3. Value Propositions

We aim to create a machine learning model that predicts future rental patterns helping companies optimize operations, resource allocation and user experience.

4. Data Acquisition

Dataset: Hourly records from Capital Bikeshare system.
https://www.kaggle.com/competitions/bike-sharing-demand/data

Data Fields:
dateime (hourly), season, holiday, workingday, weather, temperature, feel like temperature, humidity, windspeed, number of casual user, number of regresitier users, number of rentals.

5. Modeling

Initial models will include:

- Xgboost
- Random forest,
- long-short-term memory (RNN),
- light gradient boosting machine (LGBM)
- Stacking: Random forest + Xgboost
- Linear Regression Models
- Support Vector Regression

6. Model Evaluation

We will use **Root Mean Logarithmic Squared Error (RMLSE):** single numeric value that is higher when predictions systematically differ in a **logarithmic/percentage sense** from the true values, and lower when the predictions align closely in ratio.

$$\text{RMSLE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left[\log(\hat{y}_i + 1) - \log(y_i + 1)\right]^2}.$$

7. Data Preparation

Data cleaning will involve handling missing values, addressing inconsistencies in the datasets. Feature engineering will include the creation of new variables and encoding properly to enrich the dataset for better learning pattern of underly data for better prediction.

Modified from Bill Schmarzo's Machine Learning Canvas and Jasmine Vasandani's Data Science Workflow Canvas for AT85.01 Computer Programming for Data Science

Project Scope

- Prediction rental demand per hour level based on historical data using machine learning/deep learning model
- Data mining process: exploratory data analysis on single variate, bi-variate, multi-variate to understand underly pattern of data
- Data preparation & Feature engineering: filling missing data if needed, converted to right data format of features, engineer more features based on current ones
- Model selection: Splitting data strategy by using cross validation and time-based validation to prevent data leakage
- Machine learning models building:
    - **Tree-based models** (Random Forest, XGBoost, LightGBM)
    - **Linear models** (Linear, Ridge, Lasso, ElasticNet)
    - **Support Vector Machines Regression** (SVR)
    - **Deep Learning** (LSTM)
    - **Stacking:** Random Forest + SVR, Random forest + K-Nearest Neighbors
- Model evaluation metrics: R**oot Mean Squared Logarithmic Error (RMSLE), Mean Absolute Error (MAE), R2**
- Model result on test set and submit to kaggle to check the benchmark and feature important on prediction
- Deployment to web application and demo on how to use the model.

Data Set Overview

Data Fields

- **datetime** - hourly date + timestamp
- **season** - 1 = spring, 2 = summer, 3 = fall, 4 = winter
- **holiday** - whether the day is considered a holiday
- **working day** - whether the day is neither a weekend or holiday
- **weather** - 1: Clear, Few clouds, Partly cloudy, Partly cloudy
  2: Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist
  3: Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds
  4: Heavy Rain + Ice Pellets + Thunderstorm + Mist, Snow + Fog
- **temp** - temperature in Celsius
- **atemp** - "feels like" temperature in Celsius
- **humidity** - relative humidity
- **windspeed** - wind speed
- **casual** - number of non-registered user rentals initiated
- **registered** - number of registered user rentals initiated
- **count** - number of total rentals

Data Set Overview

Target of prediction

Train data format
The training set is comprised of the first 19 days of each month.

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count
0	2011-01-01 00:00:00	1	0	0	1	9.84	14.395	81	0.0000	3	13	16
1	2011-01-01 01:00:00	1	0	0	1	9.02	13.635	80	0.0000	8	32	40
2	2011-01-01 02:00:00	1	0	0	1	9.02	13.635	80	0.0000	5	27	32
3	2011-01-01 03:00:00	1	0	0	1	9.84	14.395	75	0.0000	3	10	13
4	2011-01-01 04:00:00	1	0	0	1	9.84	14.395	75	0.0000	0	1	1
...	...	...	...	...	...	...	...	...	...	...	...	...
10881	2012-12-19 19:00:00	4	0	1	1	15.58	19.695	50	26.0027	7	329	336
10882	2012-12-19 20:00:00	4	0	1	1	14.76	17.425	57	15.0013	10	231	241
10883	2012-12-19 21:00:00	4	0	1	1	13.94	15.910	61	15.0013	4	164	168
10884	2012-12-19 22:00:00	4	0	1	1	13.94	17.425	61	6.0032	12	117	129
10885	2012-12-19 23:00:00	4	0	1	1	13.12	16.665	66	8.9981	4	84	88

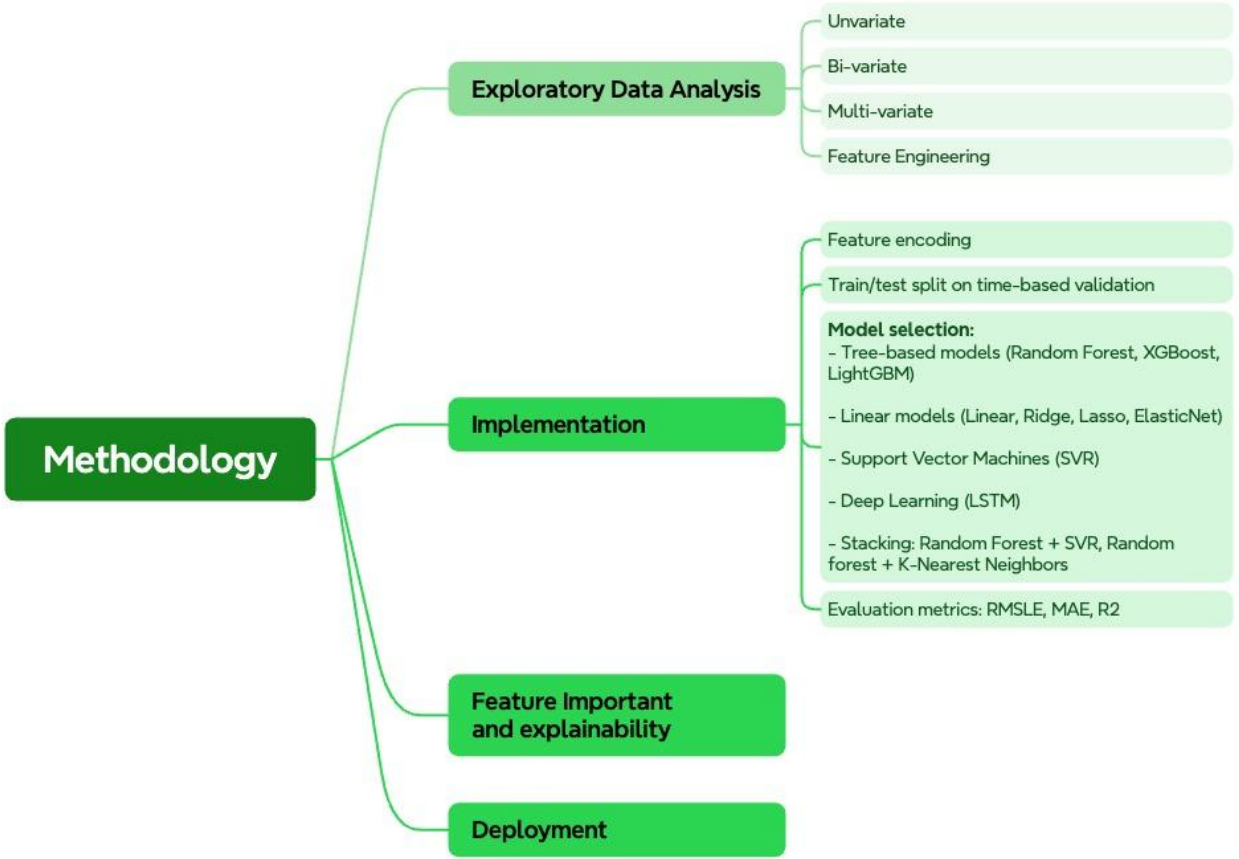
10886 rows × 12 columns

The **test data** is the 20th to the end of the month. We predict the total count of bikes rented during each hour covered by the test set, using only information available prior to the rental period.

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed
0	2011-01-20 00:00:00	1	0	1	1	10.66	11.365	56	26.0027
1	2011-01-20 01:00:00	1	0	1	1	10.66	13.635	56	0.0000
2	2011-01-20 02:00:00	1	0	1	1	10.66	13.635	56	0.0000
3	2011-01-20 03:00:00	1	0	1	1	10.66	12.880	56	11.0014
4	2011-01-20 04:00:00	1	0	1	1	10.66	12.880	56	11.0014
...	...	...	...	...	...	...	...	...	...
6488	2012-12-31 19:00:00	1	0	1	2	10.66	12.880	60	11.0014
6489	2012-12-31 20:00:00	1	0	1	2	10.66	12.880	60	11.0014
6490	2012-12-31 21:00:00	1	0	1	1	10.66	12.880	60	11.0014
6491	2012-12-31 22:00:00	1	0	1	1	10.66	13.635	56	8.9981
6492	2012-12-31 23:00:00	1	0	1	1	10.66	13.635	65	8.9981

6493 rows × 9 columns

Methodology



Feature Engineering

AIT Asian Institute of Technology

Data Cleaning

- Converted **datetime** to detailed features:
  - `Year`, `Month`, `Day`, `Hour`.

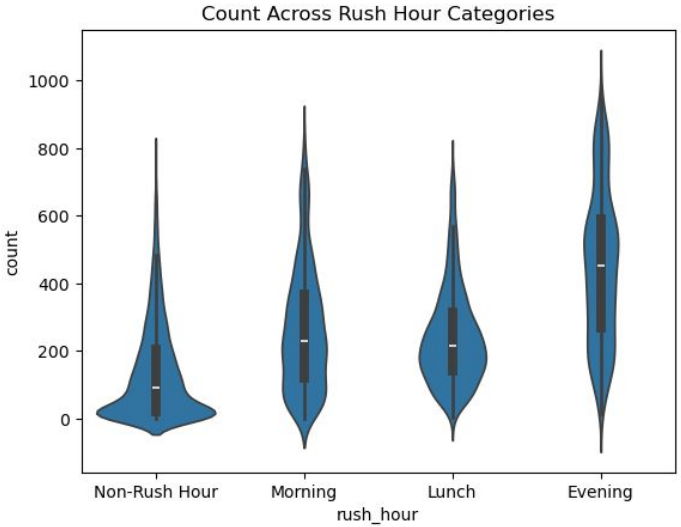
New Features Created

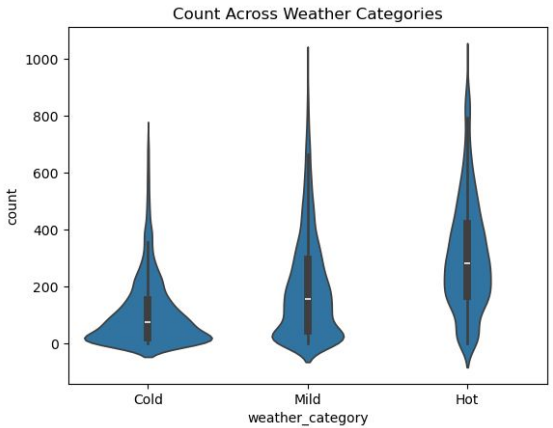
- **Time-based features:**
  - `Hour`: Captures hourly patterns.
  - `Day`, `Month`, `Year`: Captures trends over time (daily, monthly, yearly seasonality).
- **Rush Hour Classification:**
  - Identified peak commuting periods:
    - Morning rush hour (7-9 AM).
    - Lunch hour (11 AM-1 PM).
    - Evening rush hour (5-6 PM).

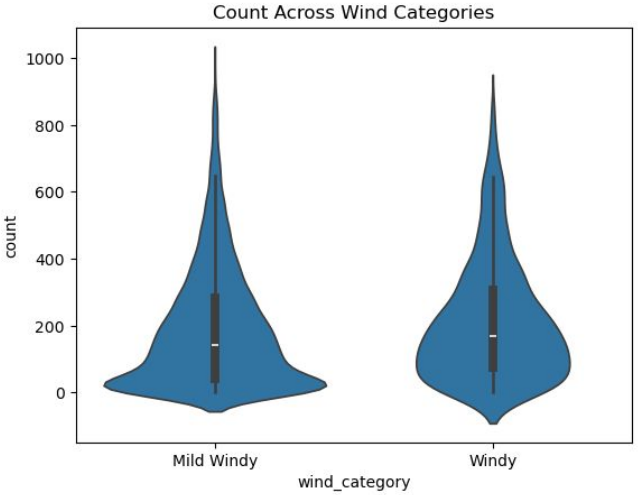
Weather-related Features

- **Temperature Categories (`weather_category`):**
  - **Hot:** > 30°C
  - **Mild:** 16-30°C
  - **Cold:** ≤ 15°C
- **Wind Speed Categories (`wind_category`):**
  - **Windy:** ≥ 25 km/h
  - **Mild Windy:** < 25 km/h

Exploratory Data Analysis

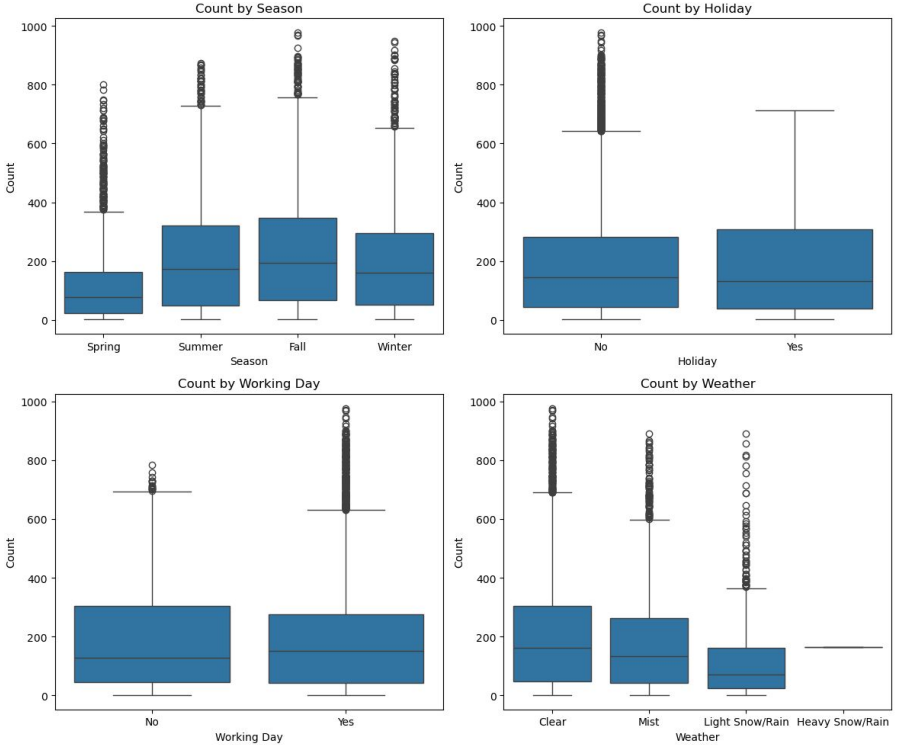






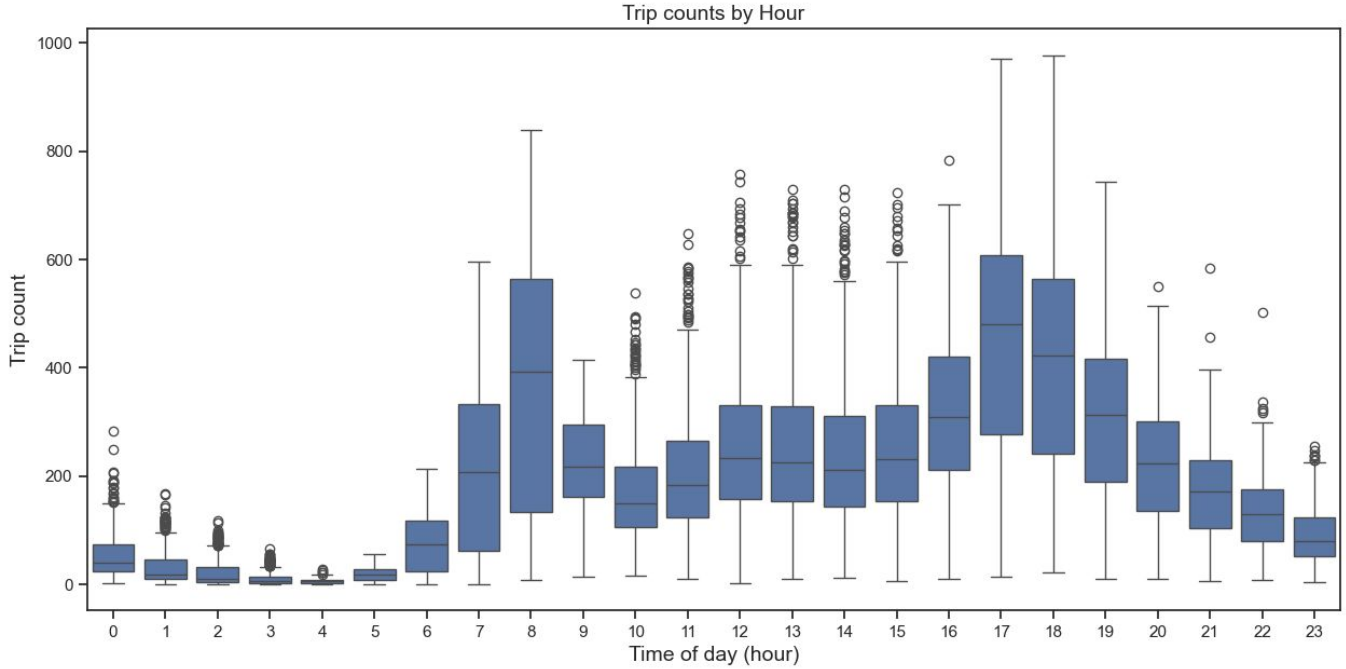
Overall, **Evening hours, Hot weather, and Windy conditions** tend to be associated with higher rental counts (or higher target values), while Non-Rush Hour, Cold weather, and lower wind speeds appear to correlate with lower counts.

Exploratory Data Analysis



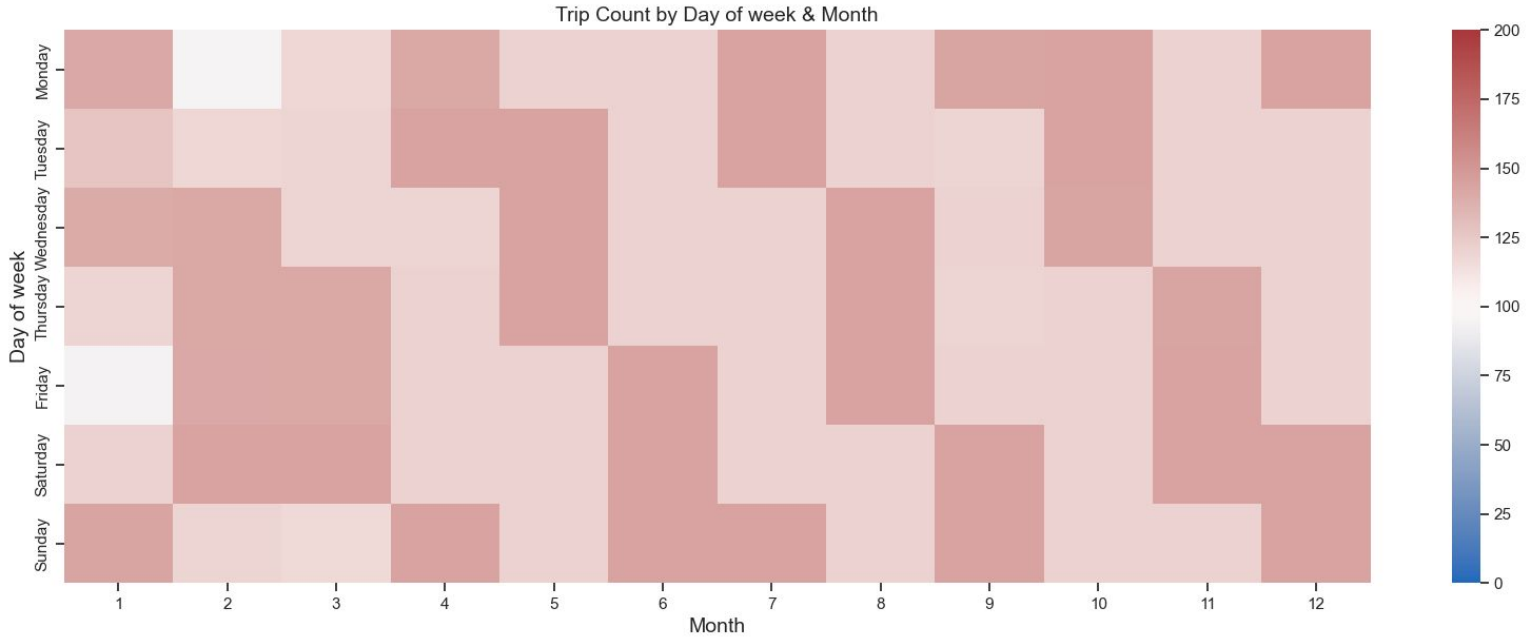
Fall, non-holiday weekdays, and clear weather are associated with higher bike rentals, while **winter, holidays, and severe weather** conditions correspond to lower usage. This indicates that seasonal and weather factors, along with day-type distinctions (holiday vs. non-holiday, weekday vs. weekend), all play a key role in explaining rental demand.

Exploratory Data Analysis



Morning and Evening Rush hour are the peak hours. Mid day hours are more outliers than the rest.

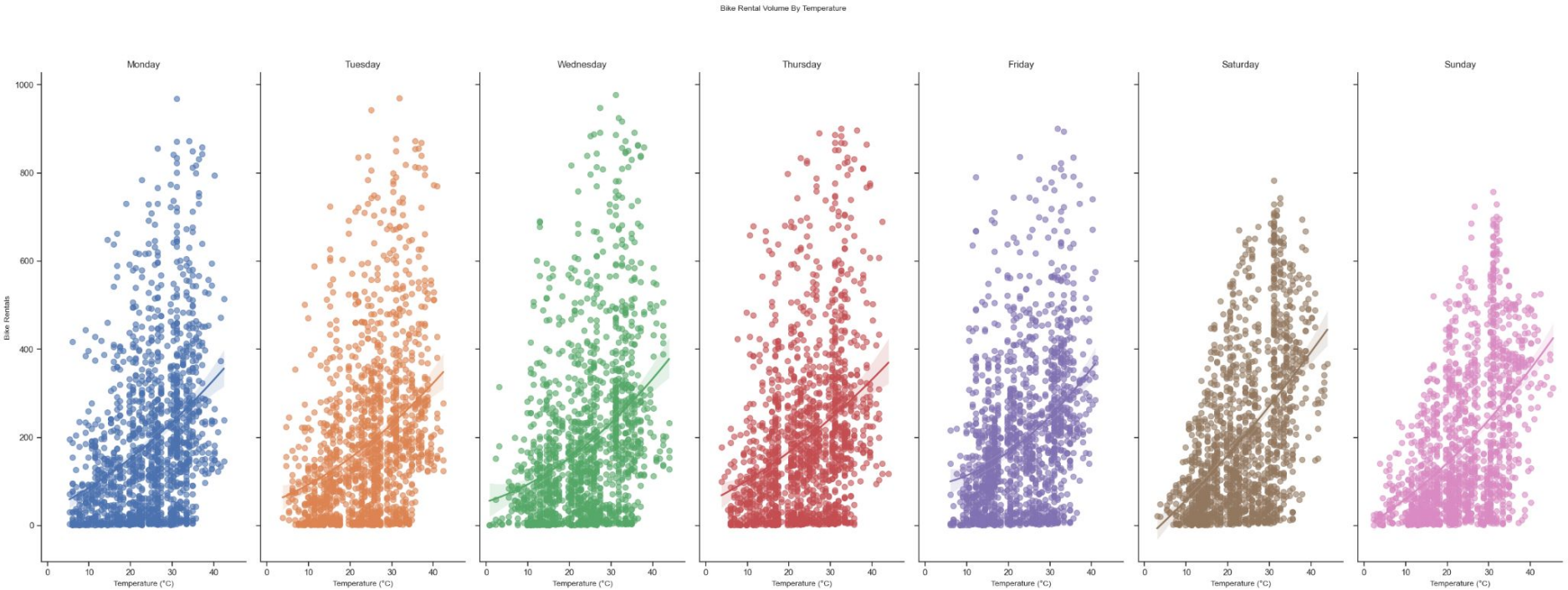
Exploratory Data Analysis



Cells more red represent higher trip counts, while lighter cells indicate fewer trips.

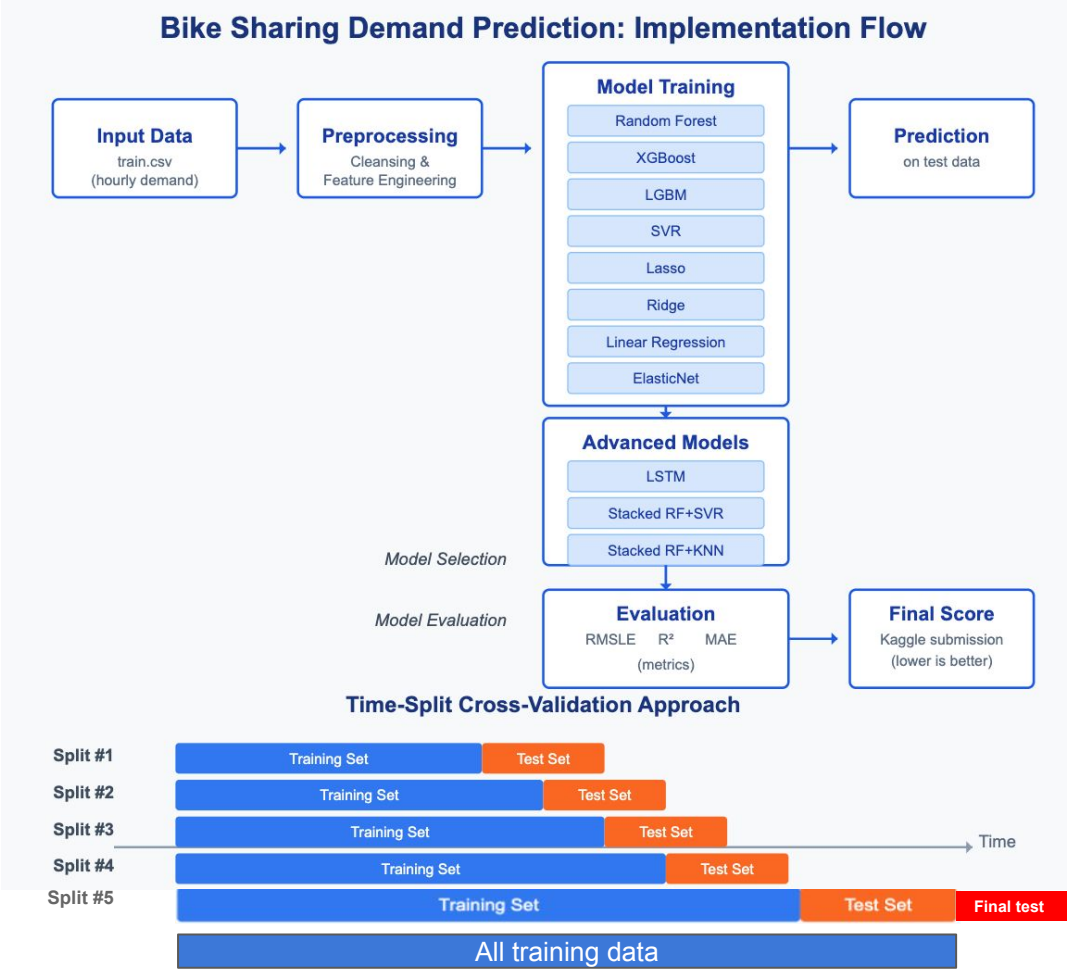
The heatmap shows that usage is not uniform across days and months; instead, it varies in a wave-like pattern.

Exploratory Data Analysis



Warmer days (especially mid-range temperatures) see more consistent demand across weekdays and weekends. Extremely cold or hot temperatures reduce usage, though weekends can still produce higher counts if conditions are otherwise favorable (e.g., sunny weekends).

Implementation



- Temporal data split prevents data leakage
- Each validation set follows its corresponding training period
- Expanding window approach uses progressively more training data
- Multiple evaluation metrics: RMSLE, R^2, and MAE for comprehensive assessment

Evaluation Metrics

$$\text{RMSLE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left[\log(\hat{y}_i + 1) - \log(y_i + 1)\right]^2}.$$

n: is the number of data in the test set

ŷ : is your predicted count

y: is the actual count

- Emphasizes relative/percentage errors.
- Less sensitive to large absolute errors for high-demand periods.
- Ideal for demand forecasting tasks (safe to zero or near zero values target variable)

Evaluation Metrics

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \widehat{y_i}|$$

where,

n: *number of observation*

y_i: *the actual value of the* i^{th} *observation*

$\widehat{y_i}$: *the predicted value of the* i^{th} *observation*

$$R^2 = 1 - \frac{\text{sum squared regression (SSR)}}{\text{total sum of squares (SST)}},$$

$$= 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}.$$

Mean Absolute Error (MAE)

- Directly interpretable in original units - "model is off by $300 on average"
- Robust to outliers - doesn't overpenalize large errors unlike MSE
- Consistent gradient for optimization - more stable training with noisy data

Coefficient of Determination (R^2)

- Normalized metric - comparable across different datasets regardless of scale
- Shows improvement over baseline - quantifies how much better than predicting the mean
- Intuitive for stakeholders - "model explains 85% of variation in the target variable"

RESULTS

Linear Models Comparison

Linear Regression

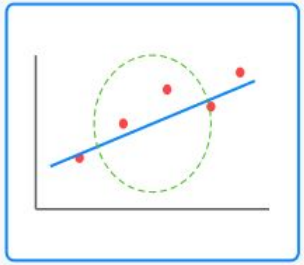
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$



Minimize: $\Sigma(y - \hat{y})^2$

Ridge Regression (L2)

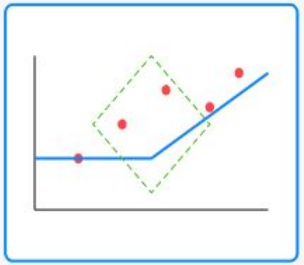
Minimize: $\Sigma(y - \hat{y})^2 + \lambda\Sigma\beta^2$



L2 Penalty (Circle)

Lasso Regression (L1)

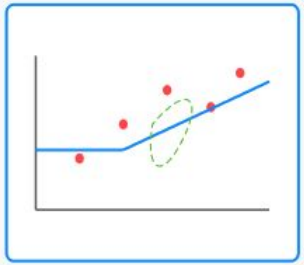
Minimize: $\Sigma(y - \hat{y})^2 + \lambda\Sigma|\beta|$



L1 Penalty (Diamond)

ElasticNet (L1 + L2)

Minimize: $\Sigma(y - \hat{y})^2 + \lambda_1\Sigma|\beta| + \lambda_2\Sigma\beta^2$



Combined Penalty

```
------------------------------------------------------------
Processing LinearRegression
------------------------------------------------------------
Training and evaluating LinearRegression...

Cross-validation metrics:
   fold    rmsle      r2       mae
0     1  1.1695  0.4270   93.4916
1     2  0.9321  0.5024   68.8404
2     3  1.2024  0.4458   88.3352
3     4  1.1886  0.5025  126.6291
4     5  1.0973  0.5070  114.1290

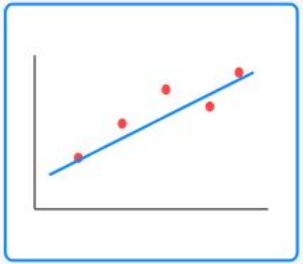
Mean metrics:
mean_rmsle: 1.1180
mean_r2: 0.4769
mean_mae: 98.2851
std_rmsle: 0.0998
std_r2: 0.0337
std_mae: 20.2157

Top 10 Most Important Features:
                Feature  Importance
rush_hour_Non-Rush Hour    110.2430
        rush_hour_Lunch     54.7118
                   Hour     49.2883
                   Year     41.4646
                  Month     32.3352
               humidity     30.7062
                   temp     27.4639
                  atemp     22.5787
               season_3     20.9505
      rush_hour_Morning     19.7345
```

Linear Models Comparison

Linear Regression

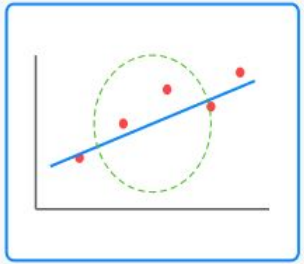
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$



Minimize: $\Sigma(y - \hat{y})^2$

Ridge Regression (L2)

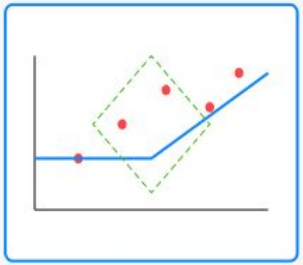
Minimize: $\Sigma(y - \hat{y})^2 + \lambda\Sigma\beta^2$



L2 Penalty (Circle)

Lasso Regression (L1)

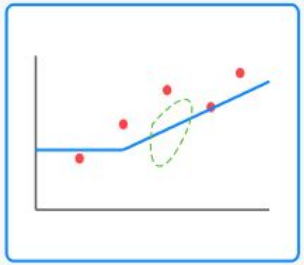
Minimize: $\Sigma(y - \hat{y})^2 + \lambda\Sigma|\beta|$



L1 Penalty (Diamond)

ElasticNet (L1 + L2)

Minimize: $\Sigma(y - \hat{y})^2 + \lambda_1\Sigma|\beta| + \lambda_2\Sigma\beta^2$



Combined Penalty

```
------------------------------------------------------------
Processing Ridge
------------------------------------------------------------
Tuning Ridge...
Fitting 5 folds for each of 5 candidates, totalling 25 fits
Best parameters: {'alpha': 0.01}
Best RMSLE from tuning: nan
Training and evaluating Ridge...

Cross-validation metrics:
   fold   rmsle      r2       mae
0     1  1.1694  0.4270   93.4901
1     2  0.9321  0.5024   68.8404
2     3  1.2024  0.4458   88.3352
3     4  1.1886  0.5025  126.6290
4     5  1.0973  0.5070  114.1290

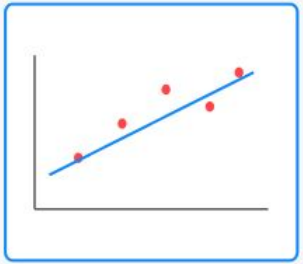
Mean metrics:
mean_rmsle: 1.1180
mean_r2: 0.4769
mean_mae: 98.2847
std_rmsle: 0.0998
std_r2: 0.0337
std_mae: 20.2158

Top 10 Most Important Features:
                  Feature  Importance
rush_hour_Non-Rush Hour    110.2426
        rush_hour_Lunch     54.7115
                   Hour     49.2883
                   Year     41.4646
                  Month     32.3343
               humidity     30.7062
                   temp     27.4637
                  atemp     22.5789
               season_3     20.9497
      rush_hour_Morning     19.7342
```

Linear Models Comparison

Linear Regression

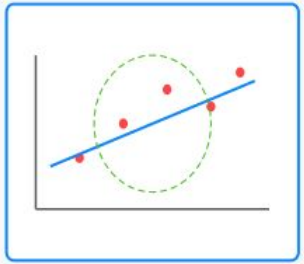
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$



Minimize: $\Sigma(y - \hat{y})^2$

Ridge Regression (L2)

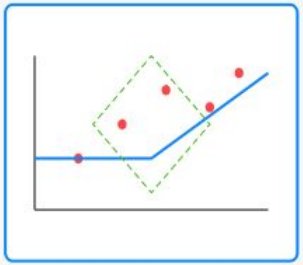
Minimize: $\Sigma(y - \hat{y})^2 + \lambda\Sigma\beta^2$



L2 Penalty (Circle)

Lasso Regression (L1)

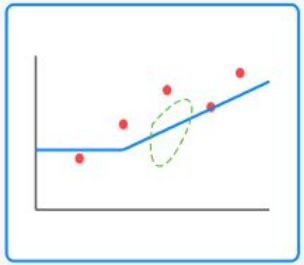
Minimize: $\Sigma(y - \hat{y})^2 + \lambda\Sigma|\beta|$



L1 Penalty (Diamond)

ElasticNet (L1 + L2)

Minimize: $\Sigma(y - \hat{y})^2 + \lambda_1\Sigma|\beta| + \lambda_2\Sigma\beta^2$



Combined Penalty

```
------------------------------------------------
Processing Lasso
------------------------------------------------
Tuning Lasso...
Fitting 5 folds for each of 5 candidates, totalling 25 fits
```

```
Best parameters: {'alpha': 0.001}
Best RMSLE from tuning: nan
Training and evaluating Lasso...

Cross-validation metrics:
   fold   rmsle      r2       mae
0     1  1.1693  0.4271   93.4810
1     2  0.9318  0.5022   68.8442
2     3  1.2025  0.4457   88.3395
3     4  1.1886  0.5026  126.6250
4     5  1.0973  0.5070  114.1291

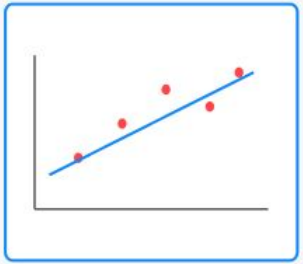
Mean metrics:
mean_rmsle: 1.1179
mean_r2: 0.4769
mean_mae: 98.2838
std_rmsle: 0.0999
std_r2: 0.0336
std_mae: 20.2136

Top 10 Most Important Features:
                  Feature  Importance
rush_hour_Non-Rush Hour    110.2365
        rush_hour_Lunch     54.7065
                   Hour     49.2887
                   Year     41.4636
                  Month     32.2884
               humidity     30.7063
                   temp     27.4749
                  atemp     22.5742
               season_3     20.9126
      rush_hour_Morning     19.7293
```

Linear Models Comparison

Linear Regression

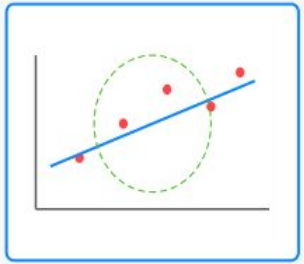
$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_n x_n$



Minimize: $\Sigma(y - \hat{y})^2$

Ridge Regression (L2)

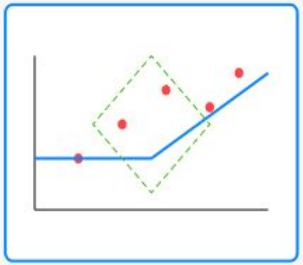
Minimize: $\Sigma(y - \hat{y})^2 + \lambda\Sigma\beta^2$



L2 Penalty (Circle)

Lasso Regression (L1)

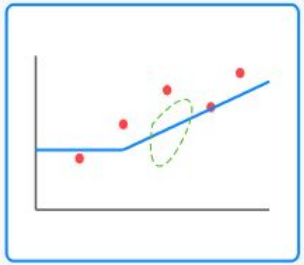
Minimize: $\Sigma(y - \hat{y})^2 + \lambda\Sigma|\beta|$



L1 Penalty (Diamond)

ElasticNet (L1 + L2)

Minimize: $\Sigma(y - \hat{y})^2 + \lambda_1\Sigma|\beta| + \lambda_2\Sigma\beta^2$



Combined Penalty

```
------------------------------------------------------------
Processing ElasticNet
------------------------------------------------------------
Tuning ElasticNet...
Fitting 5 folds for each of 25 candidates, totalling 125 fits
```

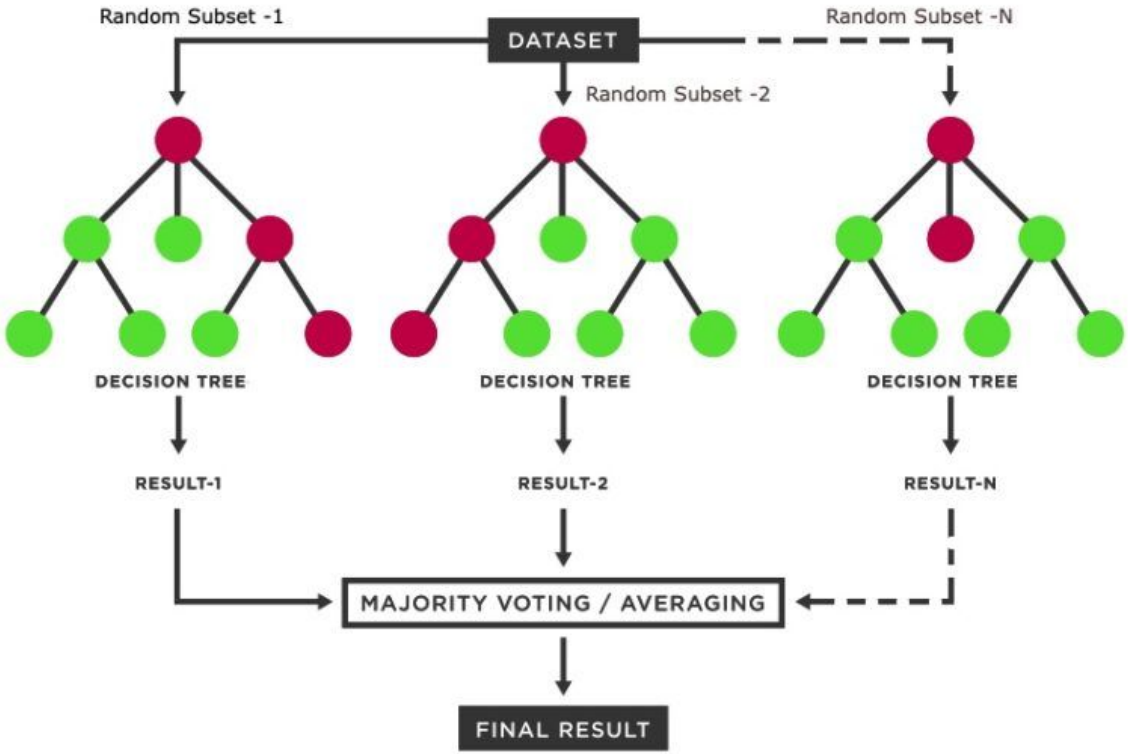
```
Best parameters: {'alpha': 10.0, 'l1_ratio': 0.9}
Best RMSLE from tuning: 1.1800
Training and evaluating ElasticNet...

Cross-validation metrics:
   fold   rmsle      r2       mae
0     1  1.1757  0.0251  110.0108
1     2  1.1858  0.3002   84.5382
2     3  1.2019  0.1740  107.1384
3     4  1.1878  0.1793  147.6344
4     5  1.1479  0.1990  137.9967

Mean metrics:
mean_rmsle: 1.1798
mean_r2: 0.1756
mean_mae: 117.4637
std_rmsle: 0.0180
std_r2: 0.0880
std_mae: 22.7074

Top 10 Most Important Features:
                Feature  Importance
rush_hour_Non-Rush Hour     30.6224
                   Hour     26.7277
                   Year     17.4188
               humidity     17.3412
                   temp     16.2473
                  atemp     16.1959
   weather_category_Hot      7.6801
                  Month      7.5087
      rush_hour_Morning      6.7224
              weather_3      2.7848
```

Random Forest model concept




```
Preparing features...
Training RandomForest model with time series cross-validation...

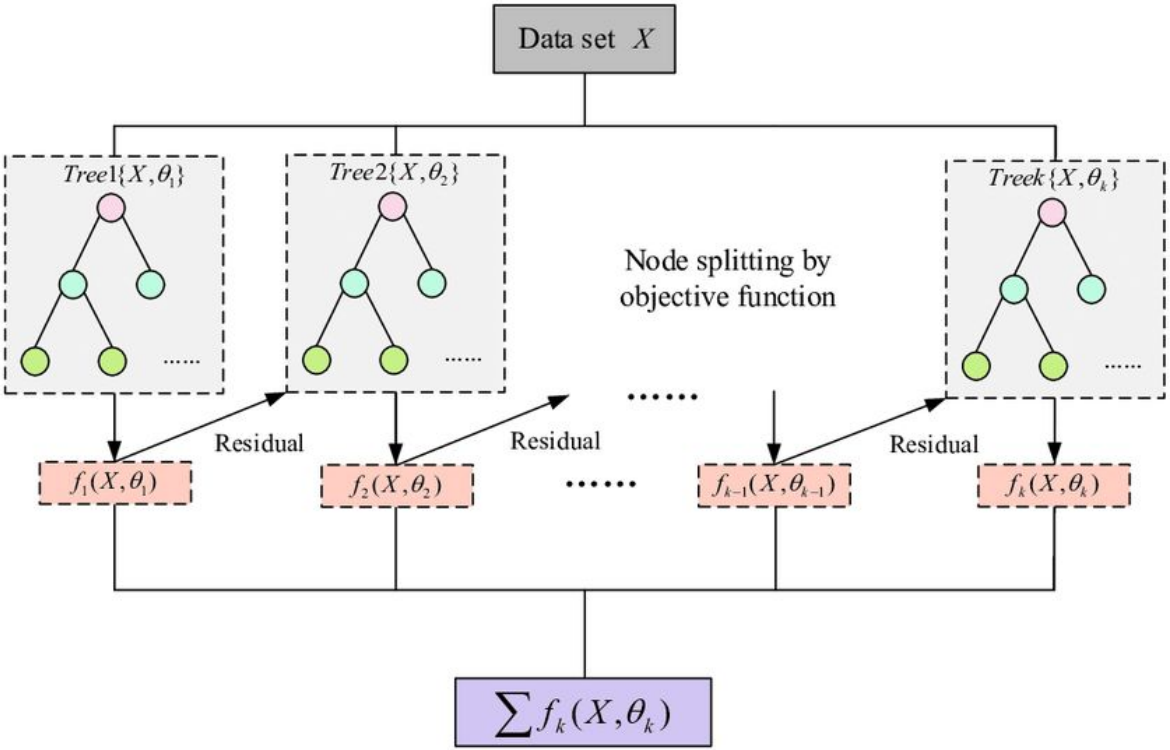
Cross-Validation Results:
   fold  train_size  test_size     rmsle        r2        mae
0     1        1816       1814  0.949796  0.609964  74.829861
1     2        3630       1814  0.504960  0.777182  42.513570
2     3        5444       1814  0.709730  0.327882  95.175955
3     4        7258       1814  0.384841  0.842126  61.761778
4     5        9072       1814  0.412904  0.792074  64.743937

Average Metrics:
Metric      Mean   Std Dev
 RMSLE  0.592446  0.236877
    R²  0.669846  0.210158
   MAE 67.805020 19.261781

Plotting feature importance...

Top 10 Most Important Features for RandomForest:
                  Feature  Importance
                     Hour    0.603561
                     Year    0.101470
                     temp    0.072307
               workingday    0.062186
                    Month    0.049362
rush_hour_Non-Rush Hour    0.043448
                    atemp    0.022934
       rush_hour_Morning    0.018306
                 humidity    0.013720
               weather_3    0.008128
```

XGBoost model concept




```
Loading and preprocessing training data...
Preparing features...
Training XGB model with time series cross-validation...

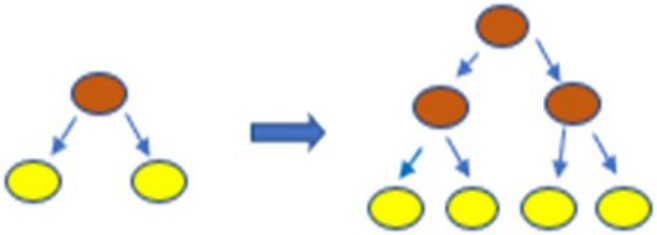
Cross-Validation Results:
 fold  train_size  test_size  rmsle     r2   mae
    1        1816       1814 1.0060 0.7022 66.00
    2        3630       1814 0.5451 0.8593 35.61
    3        5444       1814 0.8771 0.5043 85.72
    4        7258       1814 0.5661 0.8727 55.91
    5        9072       1814 0.4186 0.8779 50.79

Average Metrics:
Metric    Mean  Std Dev
 RMSLE  0.6826   0.2473
    R²  0.7633   0.1621
   MAE 58.8063  18.6232

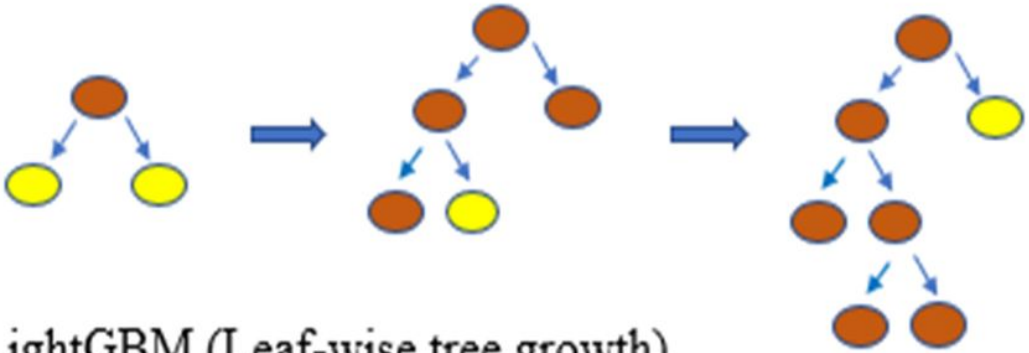
Plotting feature importance...

Top 10 Most Important Features for XGB:
                Feature  Importance
                   Hour    0.210915
                   Year    0.146206
         rush_hour_Lunch    0.136682
rush_hour_Non-Rush Hour    0.131006
             workingday    0.108324
                  atemp    0.062555
                  Month    0.050706
                   temp    0.042536
              weather_3    0.040626
               humidity    0.016117
```

LightGBM model concept



XGBoost (Level-wise tree growth)



LightGBM (Leaf-wise tree growth)

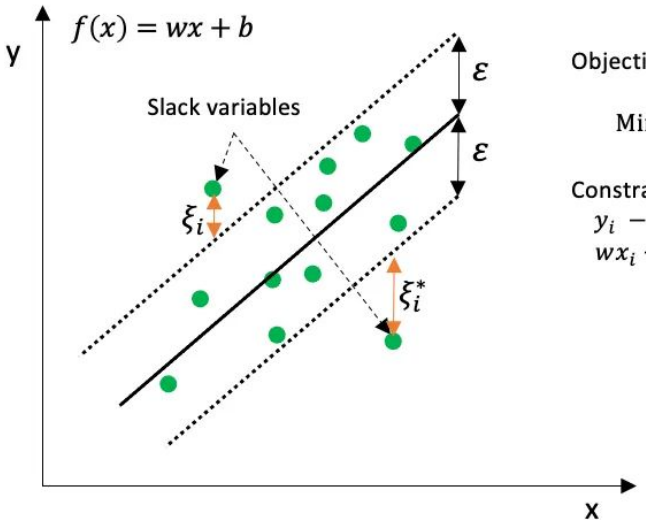
```
Cross-Validation Results:
 fold  train_size  test_size  rmsle     r2   mae
    1        1816       1814 0.6970 0.7052 59.99
    2        3630       1814 0.5582 0.8860 31.88
    3        5444       1814 0.7793 0.5174 81.88
    4        7258       1814 0.5215 0.8970 49.52
    5        9072       1814 0.4425 0.8799 50.53

Average Metrics:
Metric     Mean  Std Dev
 RMSLE  0.5997   0.1363
    R²  0.7771   0.1654
   MAE 54.7572  18.2482

Plotting feature importance...

Top 10 Most Important Features for LGBM:
                 Feature  Importance
                    Hour         822
              workingday         309
                   Month         303
                humidity         294
                     Day         247
                    temp         204
                   atemp         183
               windspeed         147
                    Year         132
rush_hour_Non-Rush Hour          90
```

Support Vector Machine Regression model concept



Univariate linear SVR (allowing for errors)

Objective:

$$\text{Minimize: } \frac{1}{2}||w||^2 + C\sum_{i=1}^{l}(\xi_i + \xi_i^*)$$

Constraints:

$$y_i - wx_i - b \leq \varepsilon + \xi_i$$
$$wx_i + b - y_i \leq \varepsilon + \xi_i^*$$
$$\xi_i^*, \xi_i^* \geq 0$$

```
Best parameters: {'C': 1.0, 'epsilon': 0.1, 'gamma': 'auto', 'kernel': 'rbf'}
Best RMSLE from tuning: 1.1973

Training and evaluating SVR with best parameters...
Training SVR model across time series folds...
Training fold 1/5...
  Fold 1 RMSLE: 1.3008, R²: -0.5303, MAE: 138.6984
Training fold 2/5...
  Fold 2 RMSLE: 1.0700, R²: 0.1619, MAE: 87.3409
Training fold 3/5...
  Fold 3 RMSLE: 1.4743, R²: -0.1745, MAE: 134.6911
Training fold 4/5...
  Fold 4 RMSLE: 1.0605, R²: 0.0693, MAE: 152.1343
Training fold 5/5...
  Fold 5 RMSLE: 1.0158, R²: 0.1256, MAE: 140.2754
Training final SVR model on all data...
Large dataset detected, using 10,000 random samples for feature importance...
Calculating feature importance using permutation method...
This may take some time depending on dataset size...

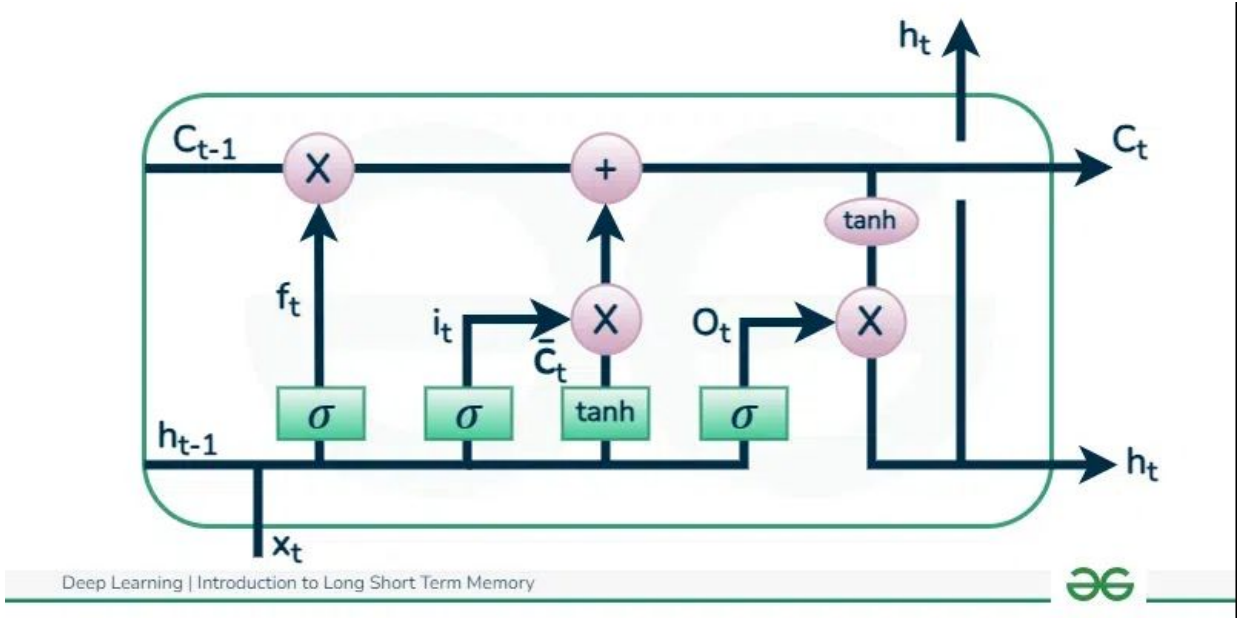
SVR Cross-validation metrics:
   fold   rmsle      r2       mae
0     1  1.3008 -0.5303  138.6984
1     2  1.0700  0.1619   87.3409
2     3  1.4743 -0.1745  134.6911
3     4  1.0605  0.0693  152.1343
4     5  1.0158  0.1256  140.2754

SVR Mean metrics:
mean_rmsle: 1.1843
mean_r2: -0.0696
mean_mae: 130.6280
std_rmsle: 0.1758
std_r2: 0.2586
std_mae: 22.4098
```

```
Top 10 Most Important Features for SVR:
                  Feature  Importance
                     Hour      0.1500
rush_hour_Non-Rush Hour      0.1316
                 humidity      0.0599
                     Year      0.0527
                     temp      0.0451
                    atemp      0.0433
        rush_hour_Morning      0.0362
               workingday      0.0226
                    Month      0.0205
   weather_category_Hot      0.0185
```

LSTM model concept

Bidirectional LSTMs (Long Short-Term Memory) are a type of recurrent neural network (RNN) architecture that processes input data in both forward and backward directions. In a traditional LSTM, the information flows only from past to future, making predictions based on the preceding context




```
==========================================
LSTM Model Analysis
==========================================

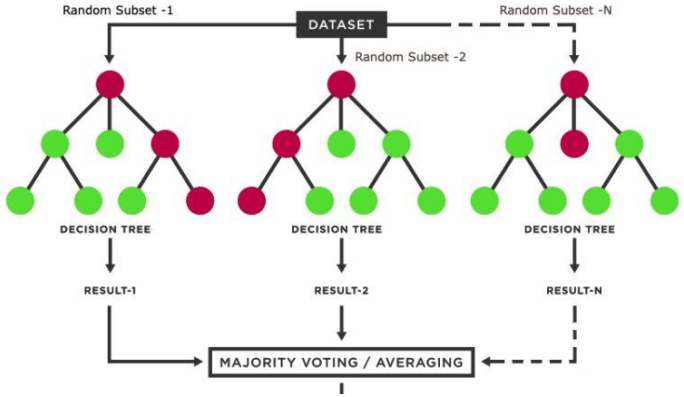
Loading and preprocessing data...
Scaling data...
Creating sequences with lookback window of 24...
X_seq shape: (10862, 24, 8), y_seq shape: (10862,)
Training samples: 7603, Testing samples: 3259

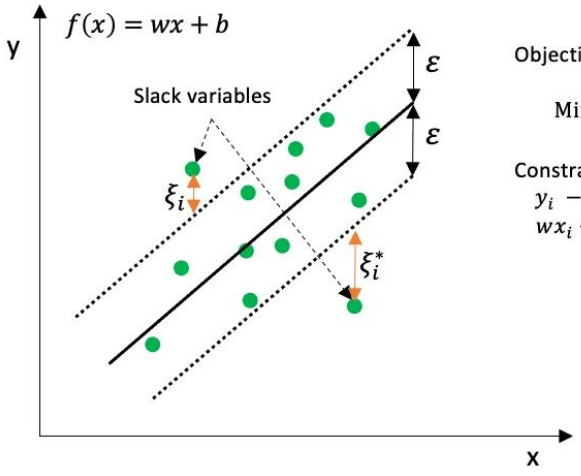
Training LSTM model:
Epoch 1/20, Loss: 0.321311
Epoch 2/20, Loss: 0.107064
Epoch 3/20, Loss: 0.073006
Epoch 4/20, Loss: 0.057016
Epoch 5/20, Loss: 0.049023
Epoch 6/20, Loss: 0.044881
Epoch 7/20, Loss: 0.040361
Epoch 8/20, Loss: 0.038028
Epoch 9/20, Loss: 0.038400
Epoch 10/20, Loss: 0.036816
Epoch 11/20, Loss: 0.035255
Epoch 12/20, Loss: 0.033775
Epoch 13/20, Loss: 0.032315
Epoch 14/20, Loss: 0.032090
Epoch 15/20, Loss: 0.031065
Epoch 16/20, Loss: 0.029247
Epoch 17/20, Loss: 0.029093
Epoch 18/20, Loss: 0.028565
Epoch 19/20, Loss: 0.026913
Epoch 20/20, Loss: 0.028244

Evaluating model...

LSTM Model Evaluation Metrics:
RMSLE: 0.3602
R²: 0.9371
MAE: 35.8887
```

Stacking Random Forest + SVR





Univariate linear SVR (allowing for errors)

Objective:

$$\text{Minimize: } \frac{1}{2}||w||^2 + C\sum_{i=1}^{l}(\xi_i + \xi_i^*)$$

Constraints:

$$y_i - wx_i - b \leq \varepsilon + \xi_i$$
$$wx_i + b - y_i \leq \varepsilon + \xi_i^*$$
$$\xi_i^*, \xi_i^* \geq 0$$

```
==========================================
Stacked Model Analysis (Random Forest + SVR)
==========================================

Loading and preprocessing training data...
Preparing features...

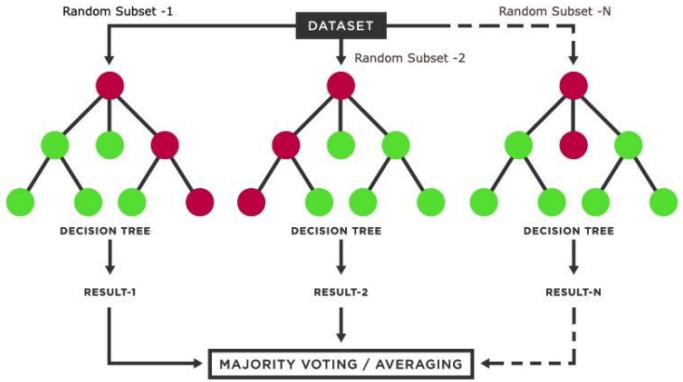
Creating stacked model (RF + SVR)...
Training stacked model across time series folds...
Training fold 1/5...
  Fold 1 RMSLE: 0.8956, R²: 0.6481, MAE: 71.0055
Training fold 2/5...
  Fold 2 RMSLE: 0.6250, R²: 0.8236, MAE: 37.9177
Training fold 3/5...
  Fold 3 RMSLE: 0.7055, R²: 0.3678, MAE: 93.9840
Training fold 4/5...
  Fold 4 RMSLE: 0.6141, R²: 0.8624, MAE: 56.0524
Training fold 5/5...
  Fold 5 RMSLE: 0.4575, R²: 0.8259, MAE: 60.1123
Training final stacked model on all data...
Calculating feature importance using permutation method...

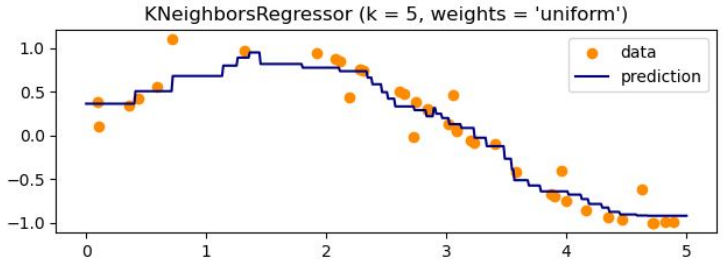
Stacked Model (RF+SVR) Cross-validation metrics:
   fold   rmsle      r2      mae
0     1  0.8956  0.6481  71.0055
1     2  0.6250  0.8236  37.9177
2     3  0.7055  0.3678  93.9840
3     4  0.6141  0.8624  56.0524
4     5  0.4575  0.8259  60.1123
```

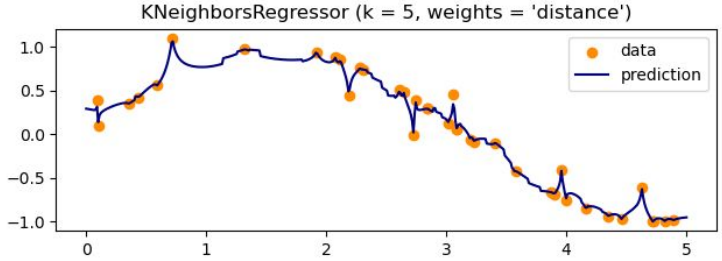
```
Stacked Model (RF+SVR) Mean metrics:
mean_rmsle: 0.6596
mean_r2: 0.7056
mean_mae: 63.8144
std_rmsle: 0.1428
std_r2: 0.1846
std_mae: 18.4743

Top 10 Most Important Features for Stacked Model (RF+SVR):
                Feature  Importance
                   Hour      1.0820
             workingday      0.2717
                   Year      0.2084
rush_hour_Non-Rush Hour      0.0851
                  Month      0.0746
                   temp      0.0700
               humidity      0.0426
                  atemp      0.0187
              weather_3      0.0183
      rush_hour_Morning      0.0103
```

Stacking Random Forest + KNN








```
Best parameters:
  RandomForest: {'max_depth': 8, 'n_estimators': 50}
  KNN: {'n_neighbors': 3, 'weights': 'uniform'}
  Meta-learner: {'alpha': 0.1}
Best RMSLE from tuning: nan

Training and evaluating stacked model...
Training stacked model across time series folds...
Training fold 1/5...
  Fold 1 RMSLE: 0.8422, R²: 0.6541, MAE: 68.3616
Training fold 2/5...
  Fold 2 RMSLE: 0.6909, R²: 0.8058, MAE: 39.3446
Training fold 3/5...
  Fold 3 RMSLE: 0.9516, R²: 0.3061, MAE: 95.1130
Training fold 4/5...
  Fold 4 RMSLE: 0.8177, R²: 0.8424, MAE: 63.5141
Training fold 5/5...
  Fold 5 RMSLE: 0.4602, R²: 0.7737, MAE: 68.4588
```

```
Stacked Model Cross-validation metrics:
   fold   rmsle      r2      mae
0     1  0.8422  0.6541  68.3616
1     2  0.6909  0.8058  39.3446
2     3  0.9516  0.3061  95.1130
3     4  0.8177  0.8424  63.5141
4     5  0.4602  0.7737  68.4588

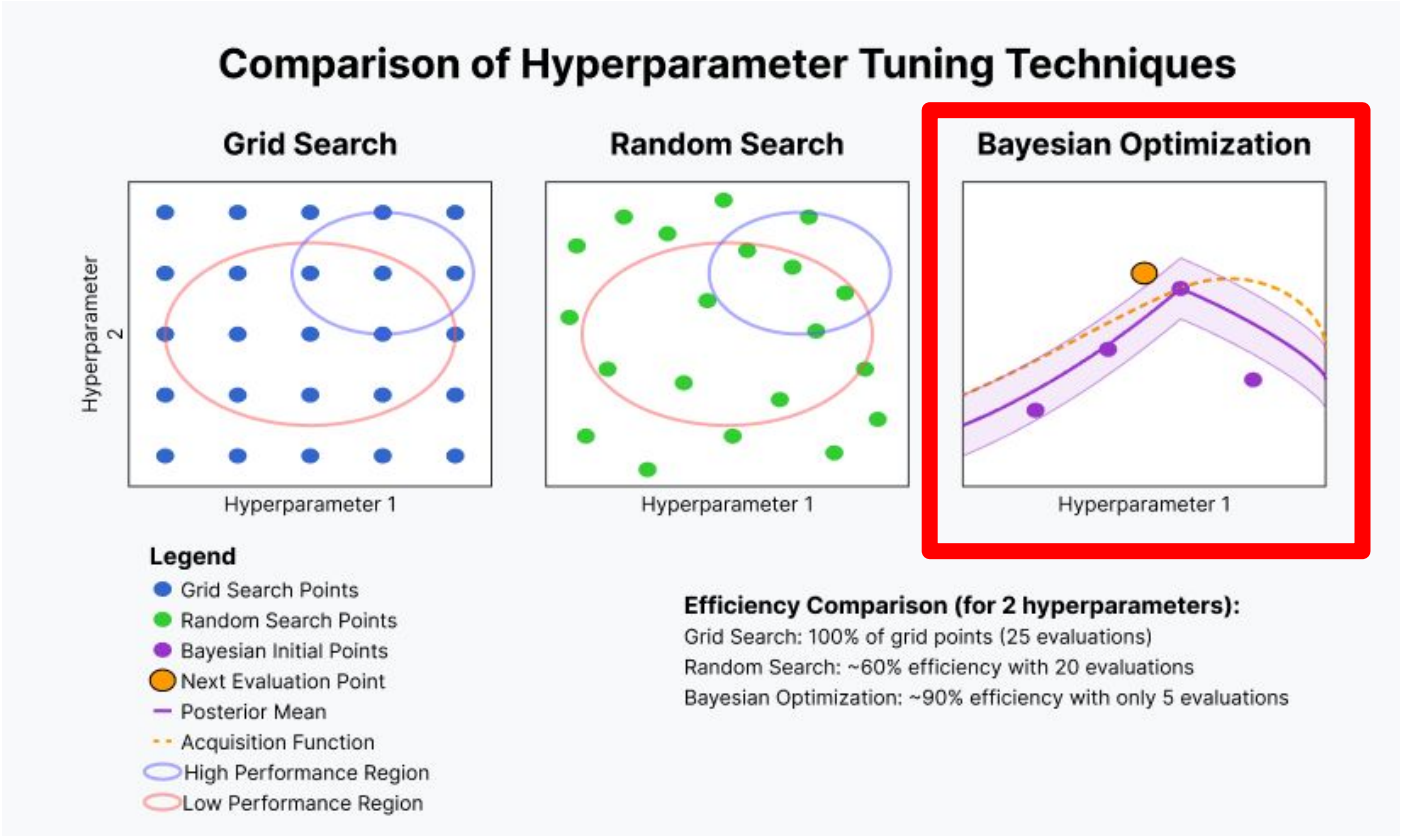
Stacked Model Mean metrics:
mean_rmsle: 0.7525
mean_r2: 0.6764
mean_mae: 66.9584
std_rmsle: 0.1680
std_r2: 0.1956
std_mae: 17.7273

Top 10 Most Important Features for Stacked Model:
                  Feature  Importance
                     Hour      0.8692
               workingday      0.2485
                     Year      0.1978
rush_hour_Non-Rush Hour      0.1023
                     temp      0.0563
                    Month      0.0459
                 humidity      0.0320
        rush_hour_Lunch      0.0225
      rush_hour_Morning      0.0214
                weather_3      0.0201
```

Model Result

Category	Model	CV RMSLE	CV R^2	CV MAE	Kaggle Score
Ensemble	Random Forest	0.5924	0.6698	67.81	0.5434
Gradient Boosting	LGBM	0.5997	0.7771	54.76	0.5461
	XGBoost	0.6826	0.7633	58.81	0.5987
Stacking	Stacked RF+SVR	0.6596	0.7056	63.81	0.8493
	Stacked RF+KNN	0.7525	0.6764	66.96	0.8925
Support Vector	SVR	1.1843	-0.0696	130.63	1.1448
Linear	Lasso	1.1179	0.4769	98.28	1.2387
	Ridge	1.118	0.4769	98.28	1.2388
	Linear Regression	1.118	0.4769	98.29	1.2388
	ElasticNet	1.1798	0.1756	117.46	1.2818
Deep Learning	LSTM	0.3602	0.9371	35.89	1.7434

Model performance improvement method



RandomForest Bayesian Optimization and Time Series Cross-Validation Results

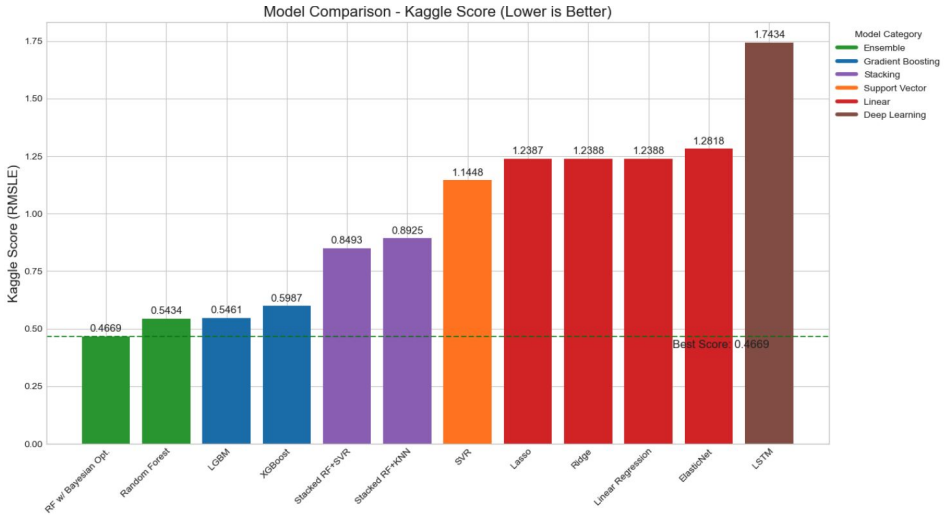
Best RMSLE: 0.51006
Total iterations: 50
Optimization time: ~1.92 hours

Parameter	Value	Description
n_estimators	500	Number of trees in the forest
max_depth	16	Maximum depth of each tree
min_samples_split	4	Minimum samples required to split a node
min_samples_leaf	10	Minimum samples required at a leaf node
min_weight_fraction_leaf	0.0	Minimum weighted fraction of samples required at a leaf node
max_features	None	Number of features to consider for best split
max_samples	0.99	Fraction of samples used for each tree
bootstrap	True	Whether bootstrap samples are used
criterion	squared_error	Function to measure split quality
min_impurity_decrease	0.0	Minimum impurity decrease required for split
ccp_alpha	0.0	Complexity parameter for pruning

Model Result

Category	Model	CV RMSLE	CV R^2	CV MAE	Kaggle Score
Ensemble	**RF w/ Bayesian Hyperparameter Optimization**	**0.51**	**0.6796**	**64.56**	**0.4669**
	Random Forest	0.5924	0.6698	67.81	0.5434
Gradient Boosting	LGBM	0.5997	0.7771	54.76	0.5461
	XGBoost	0.6826	0.7633	58.81	0.5987
Stacking	Stacked RF+SVR	0.6596	0.7056	63.81	0.8493
	Stacked RF+KNN	0.7525	0.6764	66.96	0.8925
Support Vector	SVR	1.1843	-0.0696	130.63	1.1448
Linear	Lasso	1.1179	0.4769	98.28	1.2387
	Ridge	1.118	0.4769	98.28	1.2388
	Linear Regression	1.118	0.4769	98.29	1.2388
	ElasticNet	1.1798	0.1756	117.46	1.2818
Deep Learning	LSTM	0.3602	0.9371	35.89	1.7434

Model Result



Bike Sharing Demand

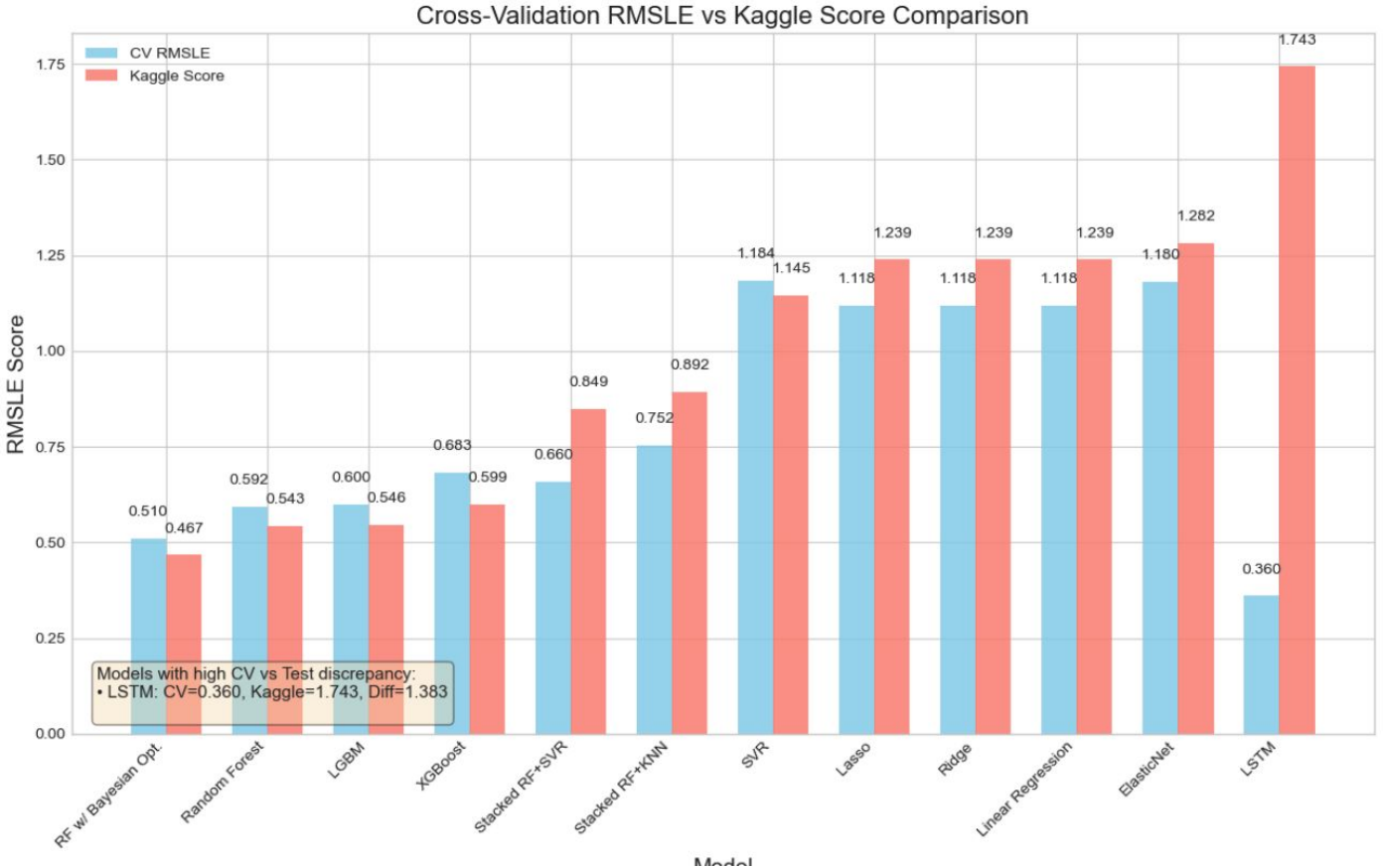
Late Submission

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

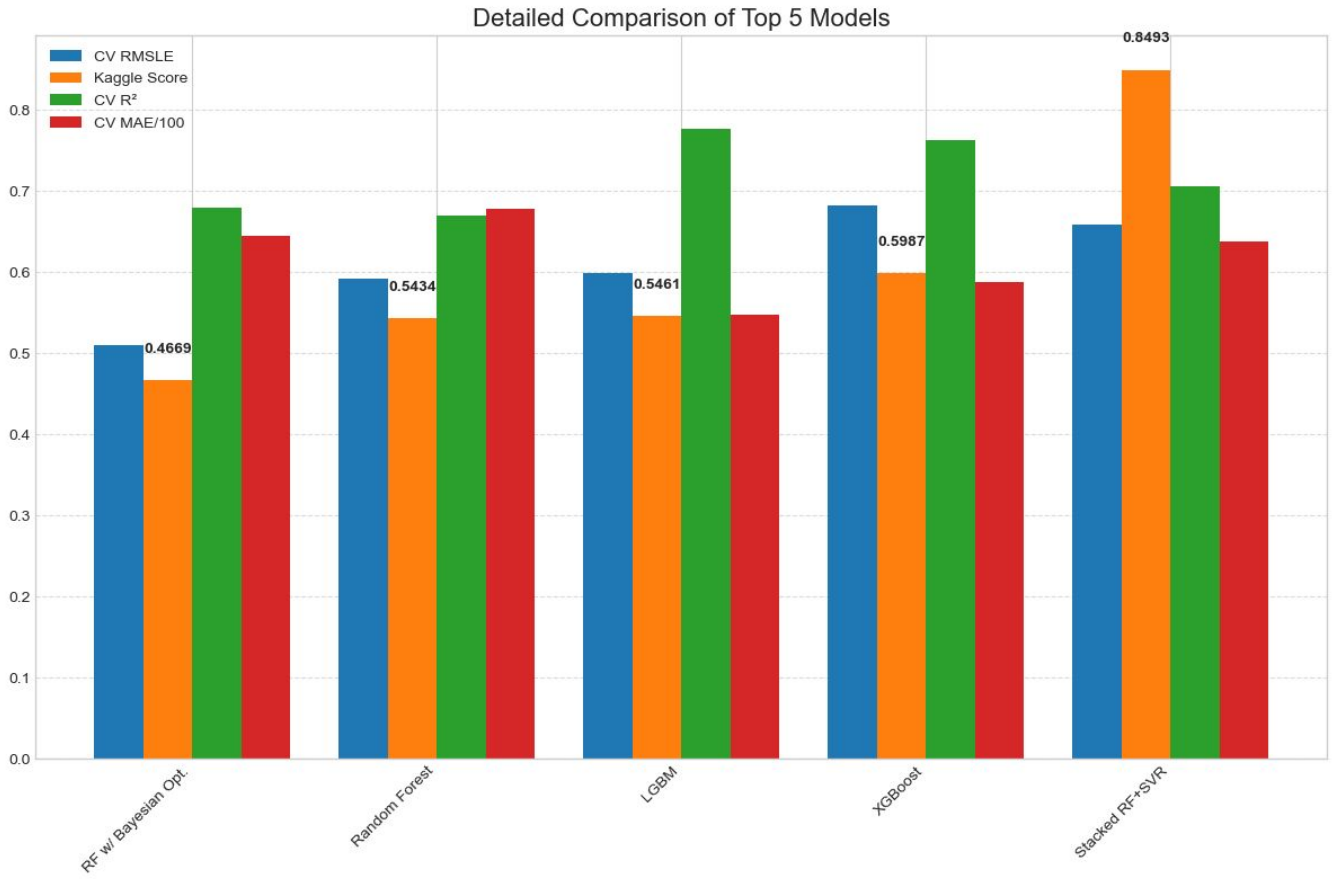
All Successful Selected Errors Recent

Submission and Description	Private Score	Public Score	Selected
randomforest_20250411_191918_submission.csv Complete (after deadline) · 8h ago	0.46690	0.46690	
stacked_rf_svr_submission.csv Complete (after deadline) · 19d ago	0.84929	0.84929	
stacked_model_submission.csv Complete (after deadline) · 19d ago	0.89253	0.89253	
svr_submission.csv Complete (after deadline) · 20d ago	1.14476	1.14476	
ridge_submission.csv Complete (after deadline) · 20d ago	1.23877	1.23877	
linearregression_submission.csv Complete (after deadline) · 20d ago	1.23878	1.23878	
elasticnet_submission.csv Complete (after deadline) · 20d ago	1.28178	1.28178	
lasso_submission.csv Complete (after deadline) · 20d ago	1.23867	1.23867	
lstm_submission.csv Complete (after deadline) · 20d ago	1.74344	1.74344	
xgb_submission.csv Complete (after deadline) · 20d ago	0.59870	0.59870	
lgbm_submission.csv Complete (after deadline) · 20d ago	0.54607	0.54607	
rf_submission_new_feature_test.csv Complete (after deadline) · 20d ago · add conformt index	0.54340	0.54340	

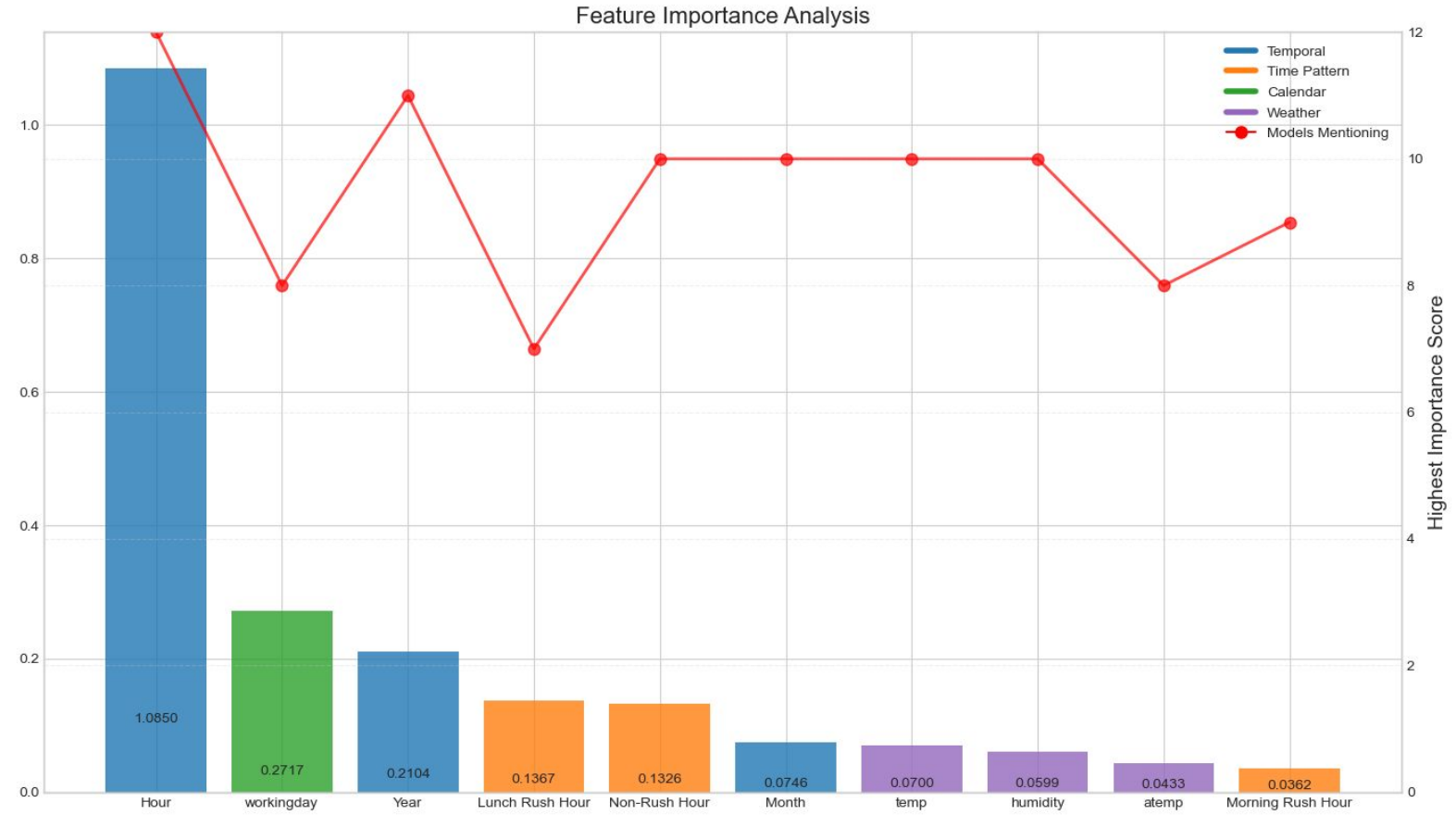
Model Result



Model Result



Feature Important and Explainability



Summary

- Tree-based models, particularly Random Forest and LightGBM, significantly outperform other model types for this task.

- Temporal features, especially hour of day and rush hour patterns, emerge as the most influential predictors across all models, highlighting the importance of time-based patterns in bike sharing demand.

- Valuable insights for bike sharing operators seeking to optimize their operations through data-driven decision-making. The high predictive accuracy of the top-performing models enables more efficient resource allocation, better rebalancing strategies, and improved customer satisfaction.

Future works

1. **Spatial-temporal modeling**: Incorporating station-level data and geographical features
2. **External data integration**: Exploring the impact of incorporating additional data sources such as public transit schedules
3. **Enhanced deep learning approaches**: Investigating more sophisticated deep learning architectures
4. **Transfer learning exploration**: Examining the transferability of trained models between different cities
5. **Interpretable machine learning**: Developing more interpretable models or explanation techniques to provide actionable insights for operational decision-making.
6. **Real-time updating**: Implementing online learning approaches that can continuously update predictions
7. **Demand simulation**: Creating simulation frameworks that can test the operational impact of different prediction models on system performance metrics.

MODEL DEMO DEPLOYMENT